Boundary and Interface Classes for Universal Chern Hamiltonians via Doubling and Jamming

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Abstract

Higson and Prodan's universal Chern Hamiltonians are closed-surface models built from the simplicial boundary operator and Poincaré duality. We treat the boundary and interface classes obtained from finite surfaces with boundary by doubling the surface and jamming the reflected copy. For every finite triangulated oriented surface with boundary we construct an intrinsic Poincaré–Lefschetz Hamiltonian, compare it with the original-side compression of the doubled closed Hamiltonian, identify the discrepancy as boundary-collar supported, and prove Schur-complement estimates for finite-M jamming. Along families with negligible collar dimension these estimates identify the normalized bulk spectral limits of the intrinsic, compressed, and jammed models. The headline Roe-algebra theorem identifies the doubled-and-jammed interface class in $K_1(C^*(Y \subset X))$ with $\partial_Y(E_+(\pi(P_\pm)))$, using the Higson–Prodan closed universal-model theorem [8, 9] for the closed-bulk gap and Chern value; this K_1-class identity is unconditional modulo HP. The integer interface winding ± 1 is proved internally in the periodic triangular Toeplitz case via the K_1-to-$\mathbb{Z}$ Toeplitz extension, and is stated as a conditional corollary for general partitions, conditional on the existence of an integer-valued Chern-jump pairing for the interface ideal. The periodic triangular Bloch class is a fully internal benchmark: its closed-bulk gap, Chern signs, and straight Toeplitz winding are proved in the paper.

Dependency categorization (sharp framing). The results split into three buckets according to which external theorems each one uses. *(A) Fully internal:* the finite-dimensional boundary theory and Schur-complement estimates, the internal bulk spectral equivalence, and the periodic triangular benchmark (gap, Chern signs, and integer Toeplitz winding for the straight periodic interface). *(B) Internal modulo Higson–Prodan:* the K_1-class identity $\mathrm{Ind}_{\mathrm{int}} = \partial_Y(E_+(\pi(P_\pm)))$ for general HP-covered boundary families. *(C) Conditional on an external integer pairing:* the integer evaluation ± 1 of this K_1-class outside the periodic Toeplitz case. The Higson–Prodan closed-bulk gap-and-Chern theorem is the *only* external theorem invoked anywhere in this paper; the Ewert–Meyer language is used to fix Mayer–Vietoris terminology only, and the required compatibility with the Roe connecting map is proved here.

Companion numerical code and the online supplement are available at the project repository (https://github.com/yspennstate/Doubling-and-Jamming-A-Universal-Chern-Hamiltonian-on-Triangulated-Surfaces-with-Boundary); additional hexagonal and square two-band symbols, robustness variants, full appendix proofs, and the numerical record are collected in that online supplement.

1 Introduction

1.1 Universal Chern Hamiltonians from simplicial data

The Haldane model is the prototypical two-dimensional class-A topological insulator [3]; for background on the tenfold-way classification and Chern phases, see [4, 5, 6, 21]. On irregular discretizations, direct magnetic-field or Peierls constructions can lead to mobility-gap or defect-state issues that complicate the extraction of clean topological gaps; see, for example, [22, 23, 24, 21]. Rigorous bulk–edge and bulk–interface correspondences in quantum Hall and related class-A systems are treated in [25, 26, 27, 30, 28, 29, 31, 32].

Higson and Prodan introduced a different route [8, 9]. For a triangulation of a *closed* oriented surface, one builds a local tight-binding Hamiltonian from two pieces of simplicial-topological data: the simplicial boundary operator B and a Hermitian Poincaré-duality operator S constructed from cap product with the fundamental cycle. The resulting Hamiltonians

$$H_{\pm} = (B + B^{\dagger}) \pm S \tag{1.1}$$

are local and, in the sequential-refinement limit, define nontrivial topological bulk classes with Chern numbers ± 1 [8, 9]. The underlying Hilbert–Poincaré framework uses the algebraic identity $BS + SB^{\dagger} = 0$, rooted in analytic surgery theory and the work of Higson and Roe [10, 11, 12, 13].

1.2 The boundary case

If the input triangulation $\mathcal{K}$ has boundary, there is no closed fundamental 2-cycle supported on the faces of $\mathcal{K}$, so the closed-case cap-product recipe for S is not directly available. Compact manifolds with boundary instead carry a relative fundamental class and satisfy Poincaré–Lefschetz duality [15]. We use this relative class to define an intrinsic boundary Hamiltonian on $\mathcal{K}$, and then compare it with a second construction that reduces the boundary problem to the closed case by simplicial doubling and jamming the reflected side.

1.3 Doubling and jamming

Given a triangulated surface-with-boundary $\mathcal{K}$, let $\overline{\mathcal{K}}$ be an orientation-reversed copy. Gluing $\mathcal{K}$ and $\overline{\mathcal{K}}$ along $\partial\mathcal{K}$ produces the closed double $D(\mathcal{K})$. We then build the closed universal Hamiltonians on $D(\mathcal{K})$ and suppress the reflected copy by adding a large on-site potential on the copy degrees of freedom. This is the *doubling-and-jamming* construction. The finite-M analysis gives error bounds for this approximation. For refinement families whose closed doubles satisfy the hypotheses of the Higson–Prodan closed-bulk theorem (Theorem 4.2; hypothesis check for the doubled families in Theorem 4.3), that theorem supplies the closed-bulk gap and Chern class. Section 4 proves, for the original/copy partition, that the Roe connecting map used to define the interface class is the Ewert–Meyer bulk–interface map for the doubled-jammed data.

1.4 Theorem map

Let

$$H_{\pm,M} = H_{\pm}(D(\mathcal{K})) + M\mathbf{P}_{\text{copy}}$$

be the jammed Hamiltonian, written in block form as

$$H_{\pm,M} = \begin{pmatrix} A & C \\ C^{\dagger} & D + M\mathbf{1} \end{pmatrix} \quad \text{on} \quad \mathcal{H}_{\text{orig}} \oplus \mathcal{H}_{\text{copy}}.$$

1. The relative fundamental class of $(\mathcal{K}, \partial\mathcal{K})$ gives an intrinsic Poincaré–Lefschetz boundary Hamiltonian on $\mathcal{K}$. The simplicial double gives a second Hamiltonian by closed Higson–Prodan theory; its original-side compression agrees with the relative Hamiltonian up to a boundary-collar operator, and the jammed operator converges to this compression at low energy.

2. The finite-M comparison is made quantitative by Schur-complement estimates: compressed resolvents, full block resolvents, first-order effective operators, bounded-energy spectral localization, leakage, Morse-index stability, and contour spectral projections. Combined with the collar-rank comparison, these estimates give a fully internal bulk spectral-measure equivalence between the intrinsic Poincaré–Lefschetz model, the doubled compression, and the finite-M jammed model along any family with negligible boundary-collar dimension and $M \to \infty$.

3. For refinement families whose closed doubles satisfy the hypotheses of Theorem 4.2 (verified in Theorem 4.3), the HP gap gives the quotient by the interface ideal and hence the global Roe-algebra interface class

$$\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M}; T^{+}, T^{-}) = \partial_Y\big(E_+(\pi(P_\pm))\big) \in K_1(C^*(Y \subset X)).$$

   Theorem 4.9, proved in Section 4, identifies this class as $\partial_Y(E_+(\pi(P_\pm)))$. In the straight periodic Toeplitz case the integer pairing is internal (Corollary 4.10) and gives winding ± 1; for general partitions the integer pairing is recorded as a conditional corollary (Corollary 4.11).

4. For the periodic triangular-lattice class, the closed-bulk gap and Chern input are checked directly. We write down the 6×6 Bloch symbol, prove a uniform zero gap from an exact determinant formula, compute the Chern signs from the isolated massless crossing, and evaluate primitive straight periodic interfaces by the Toeplitz extension. Finite covers, annular families, finite-corner quotients, finite-defect variants, and tail-split component maps are treated in the online supplement, Section 6.4.

Logical dependencies. The proof uses the following implications; no converse is assumed.

(I)	internal finite-dimensional results: the relative boundary operator, the collar comparison, and the large-M Schur-complement estimates;
(HP)	the Higson–Prodan closed-bulk gap and Chern theorem for the doubles;
(EM)	the internal identification of the original/copy Roe boundary map with the Ewert–Meyer pairing (Theorem 4.9).

From (I) and the gap part of (HP), the Roe construction supplies the global interface K_1-class. From the Chern part of (HP) and (EM), Theorem 4.9 identifies the K_1-class as $\partial_Y(E_+(\pi(P_\pm)))$. Thus the headline conclusion of Theorem 4.9 is the unconditional K_1-class identity (I)+(HP)+(EM) $\Longrightarrow$ $\mathrm{Ind}_{\mathrm{int}} = \partial_Y(E_+(\pi(P_\pm)))$. The integer evaluation $\mathrm{Wind} = \pm 1$ is internal in the periodic triangular Toeplitz case (Corollary 4.10) and conditional in general (Corollary 4.11). The ingredients (I) and (EM) are proved here. The ingredient (HP) is exactly the cited closed theorem of Higson–Prodan. In the periodic triangular section, the gap and Chern signs are proved directly, and primitive straight periodic interfaces are evaluated internally by the Toeplitz extension.

Table 1: Theorem map. "Internal" means proved here from the inputs shown; "modulo HP" means that the only cited closed-bulk theorem used is Higson–Prodan's published PRL/arXiv theorem.

Category and conclusion	Inputs	Status
Finite-dimensional boundary theory: relative boundary Hamiltonian and collar comparison	Finite triangulated surface with full boundary subcomplex	Internal, Theorem 2.8
Finite-dimensional jamming theory: finite-M decoupling, leakage, contour projections, Morse index	Finite block separation hypotheses	Internal, Section 3
Internal bulk spectral measures: PL model, doubled compression, and finite-M jammed model have the same normalized bulk spectral limits	Uniform finite-dimensional bounds, $M \to \infty$, and boundary-collar dimension $o(\dim \mathcal{H}(\mathcal{K}_n))$	Internal, Theorem 3.16
Roe-algebra K_1-class identity: global interface $\partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X))$	The Higson–Prodan closed-bulk gap/Chern theorem and the internal original/copy Roe/Ewert–Meyer identification	Internal modulo HP, Theorem 4.9
Toeplitz integer winding ± 1 for straight periodic interfaces	Periodic triangular gap, Chern signs, BDF–Toeplitz extension	Internal, Theorems 4.10 and 5.7
Integer winding for general partitions	Hypothesized Chern-jump-compatible integer pairing on $K_1(\mathcal{I})$	Conditional on the pairing's existence, Theorem 4.11
Internal periodic examples: triangular closed-bulk input	Periodic triangular Bloch symbol	Internal gap and Chern signs, Theorems 5.2 and 5.5

Table 2: Notation used for the bulk, jammed, and sidewise interface operators.

Symbol	Meaning
$H_\pm = (B + B^\dagger) \pm S$	Closed universal Chern Hamiltonian.
$H_{\pm,M} = H_\pm(D(\mathcal{K})) + M\mathbf{P}_{\mathrm{copy}}$	Finite doubled-and-jammed Hamiltonian.
$A = P_{\mathrm{orig}} H_\pm(D(\mathcal{K}))P_{\mathrm{orig}}$	Original-side compression of the doubled Hamiltonian.
$H_\pm^{\mathrm{PL}}(\mathcal{K})$	Intrinsic Poincaré–Lefschetz boundary Hamiltonian.
$P_\pm = \chi_{(-\infty,0)}(H_\pm^\infty)$	Closed-bulk Fermi projection in the coarse disjoint union.
$J_{\pm,M}^\infty$	Direct-sum jammed family defining the interface.
T^+, T^-	Positive-side and negative-side comparison operators.
Y, $C^*(Y \subset X)$	Coarse interface and its Roe ideal.
$\partial_Y(E_+(\pi(P_\pm)))$	Internally constructed interface K_1-class.

2 Simplicial setup and universal operators

Definition 2.1 (Triangulated surface with boundary)**.** A finite oriented 2-dimensional simplicial complex $\mathcal{K}$ is a *triangulated surface with boundary* if its geometric realization $|\mathcal{K}|$ is a compact PL 2-manifold with boundary and the boundary subcomplex $\partial\mathcal{K}$ is a *full subcomplex*: every simplex of $\mathcal{K}$ whose vertices all lie on $\partial\mathcal{K}$ is itself a boundary simplex. Every edge is incident to exactly one

or two faces; the edges incident to exactly one face form $\partial\mathcal{K}$, and $\partial\mathcal{K}$ is a disjoint union of cycles.

The fullness condition is mild: it holds automatically after one barycentric subdivision. Its role in keeping the simplicial double an abstract simplicial complex (without duplicate simplices or boundary-edge valence failures) is discussed in detail in the supplement, Section 2.

We work on the simplicial Hilbert space $\mathcal{H}(\mathcal{K}) = \bigoplus_{p=0}^{2} \ell^2(\mathcal{K}_p)$ with the canonical orthonormal basis indexed by oriented simplices. The simplicial boundary operator B is the standard chain boundary, viewed as a degree-lowering operator. Then $B^2 = 0$ and $B + B^\dagger$ is self-adjoint and local. We endow the set of barycenters $X(\mathcal{K})$ with the graph metric on the 1-skeleton of the first barycentric subdivision; an operator T has *propagation at most* R when $\langle \sigma, T\tau \rangle = 0$ for $\operatorname{dist}(\sigma, \tau) > R$.

For a *closed* oriented triangulation $\mathcal{K}$, cap product with the fundamental cycle $|X\rangle = \sum_{\tau \in \mathcal{K}_2} |\tau\rangle$ gives the Higson–Prodan operator P from which one defines a Hermitian, degree-reversing, local operator S by the degree-dependent symmetrization

$$T|_{\mathcal{H}_p} = \tfrac{1}{2}\big(P^\dagger + (-1)^p P\big), \qquad S|_{\mathcal{H}_p} = i^{p(p-1)+1} T|_{\mathcal{H}_p},$$

so that $BS + SB^\dagger = 0$. The closed universal Chern Hamiltonian is $H_\pm(\mathcal{K}) = (B + B^\dagger) \pm S$.

2.1 Closed-case gap criteria

Two finite-dimensional gap criteria are used as building blocks; their proofs are short and collected in the supplement.

Proposition 2.2 (Exact gap lower bound under exact anticommutation)**.** *Let $\mathcal{K}$ be closed and assume $(B+B^\dagger)S + S(B+B^\dagger) = 0$. Then $H_\pm(\mathcal{K})^2 = (B+B^\dagger)^2 + S^2$. If $\mu_S := \operatorname{dist}(0, \operatorname{spec}(S)) > 0$, then $\operatorname{spec}(H_\pm(\mathcal{K})) \cap (-\mu_S, \mu_S) = \emptyset$.*

Proof. See the online supplement, Section 6.10.

Proposition 2.3 (Gap lower bound with anticommutation defect)**.** *With $Q = B + B^\dagger$, $\mu_S := \operatorname{dist}(0, \operatorname{spec}(S))$, and $\varepsilon_{\mathrm{ac}} := \|QS + SQ\|$, if S is Hermitian then $H_\pm(\mathcal{K})^2 \succeq (\mu_S^2 - \varepsilon_{\mathrm{ac}})\mathbf{1}$.*

Proof. See the online supplement, Section 6.11.

2.2 Doubling, jamming, and the boundary Hamiltonians

Given $\mathcal{K}$ with boundary in the sense of Definition 2.1, let $\overline{\mathcal{K}}$ be an oriented copy with reversed face orientation, sharing each boundary vertex of $\mathcal{K}$. The *double* $D(\mathcal{K}) = \mathcal{K} \cup_{\partial\mathcal{K}} \overline{\mathcal{K}}$ is a finite closed oriented triangulated surface:

Proposition 2.4 (The double is a closed oriented triangulated surface)**.** *Let $\mathcal{K}$ satisfy Definition 2.1. Then $D(\mathcal{K})$ is a finite closed oriented abstract simplicial complex that triangulates a closed surface.*

Proof. See the online supplement, Section 6.12.

Theorem 2.5 (Simplex count identities under doubling)**.** $|D(\mathcal{K})_2| = 2|\mathcal{K}_2|$, $|D(\mathcal{K})_1| = 2|\mathcal{K}_1| - |\partial\mathcal{K}_1|$, $|D(\mathcal{K})_0| = 2|\mathcal{K}_0| - |\partial\mathcal{K}_0|$, *and* $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. *Consequently* $\dim \mathcal{H}(D(\mathcal{K})) = 2 \dim \mathcal{H}(\mathcal{K}) - 2|\partial\mathcal{K}_0|$, $\chi(D(\mathcal{K})) = 2\chi(\mathcal{K})$, *and for a connected, orientable, genus-g surface with $b \geq 1$ boundary components, the doubled genus is* $g_{\mathrm{double}} = 2g + b - 1$.

Proof. See the online supplement, Section 6.13.

The closed-case construction therefore produces the operator $H_\pm(D(\mathcal{K}))$ on $\mathcal{H}(D(\mathcal{K}))$. We split

$$\mathcal{H}(D(\mathcal{K})) = \mathcal{H}_{\mathrm{orig}} \oplus \mathcal{H}_{\mathrm{copy}}$$

and *jam* the reflected copy by the on-site potential

$$H_{\pm,M} = H_\pm(D(\mathcal{K})) + M\mathbf{P}_{\mathrm{copy}}. \tag{2.1}$$

The block decomposition $H_\pm(D(\mathcal{K})) = \begin{pmatrix} A & C \\ C^\dagger & D \end{pmatrix}$ on $\mathcal{H}_{\mathrm{orig}} \oplus \mathcal{H}_{\mathrm{copy}}$ defines the *original-side boundary Hamiltonian* $H^\partial_\pm(\mathcal{K}) := A$, a self-adjoint local operator on the surface-with-boundary.

The intrinsic alternative uses the relative chain complex $C_p(\mathcal{K}, \partial\mathcal{K}) = C_p(\mathcal{K})/C_p(\partial\mathcal{K})$ and the relative fundamental chain $[\mathcal{K}, \partial\mathcal{K}] = \sum_{\tau \in \mathcal{K}_2} |\tau\rangle$. Cap product with $[\mathcal{K}, \partial\mathcal{K}]$, followed by the canonical lift $\widehat{P}_p := j_{2-p} P^{\mathrm{rel}}_p : C_p(\mathcal{K}) \to C_{2-p}(\mathcal{K})$, gives the Hermitian, degree-reversing, local relative duality operator $S^{\mathrm{PL}}_{\mathcal{K}}$ on $\mathcal{H}(\mathcal{K})$. The *relative Poincaré–Lefschetz boundary Hamiltonian* is

$$H^{\mathrm{PL}}_\pm(\mathcal{K}) := B_{\mathcal{K}} + B^\dagger_{\mathcal{K}} \pm S^{\mathrm{PL}}_{\mathcal{K}} \quad \text{on } \mathcal{H}(\mathcal{K}). \tag{2.2}$$

Lemma 2.6 (Relative cap identity and collar defect)**.** *With $\partial^{\mathrm{rel}}_r : C_r(\mathcal{K}, \partial\mathcal{K}) \to C_{r-1}(\mathcal{K}, \partial\mathcal{K})$ defined by $\partial^{\mathrm{rel}}_r q_r = q_{r-1} B_{\mathcal{K},r}$, and $\delta_p := B^\dagger_{\mathcal{K},p+1}$, the relative cap-product blocks satisfy $\partial^{\mathrm{rel}}_{2-p} P^{\mathrm{rel}}_p + (-1)^p P^{\mathrm{rel}}_{p+1} \delta_p = 0$ for $p = 0, 1$. After lifting by j, the difference $B_{\mathcal{K},2-p}\widehat{P}_p + (-1)^p \widehat{P}_{p+1}\delta_p$ has matrix coefficients only when both source and target lie in a fixed barycentric collar of $\partial\mathcal{K}$. Consequently the relative duality defect $B_{\mathcal{K}} S^{\mathrm{PL}}_{\mathcal{K}} + S^{\mathrm{PL}}_{\mathcal{K}} B^\dagger_{\mathcal{K}}$ is supported in a fixed boundary collar.*

Proof. See the online supplement, Section 6.17.

Lemma 2.7 (Coefficient agreement off the boundary collar)**.** *Let P_D be the closed cap-product operator on $D(\mathcal{K})$ and $\widehat{P}$ the lifted relative cap-product operator on $\mathcal{K}$. There is a radius $R_\frown$ depending only on the local cap-product rule, such that $\langle \sigma, P_D \tau \rangle = \langle \sigma, \widehat{P}\tau \rangle$ whenever $\sigma, \tau \in \mathcal{K} \setminus N_{R_\frown}(\partial\mathcal{K})$.*

Proof. See the online supplement, Section 6.18.

Theorem 2.8 (Relative boundary Hamiltonian and doubled compression)**.** *Let $\mathcal{K}$ be a triangulated oriented surface with full boundary subcomplex. Under the natural identification of $\mathcal{H}(\mathcal{K})$ with $\mathcal{H}_{\mathrm{orig}} \subset \mathcal{H}(D(\mathcal{K}))$, $H^{\mathrm{PL}}_\pm(\mathcal{K})$ is self-adjoint and local. Moreover there is a self-adjoint operator $R_{\partial,\pm}$ such that*

$$H^\partial_\pm(\mathcal{K}) = H^{\mathrm{PL}}_\pm(\mathcal{K}) + R_{\partial,\pm}, \tag{2.3}$$

and the comparison error $R_{\partial,\pm}$, and also the relative duality defect $B_{\mathcal{K}} S^{\mathrm{PL}}_{\mathcal{K}} + S^{\mathrm{PL}}_{\mathcal{K}} B^\dagger_{\mathcal{K}}$, are supported in a bounded collar of the boundary: there is a radius R_{PL}, depending only on the local cap-product convention, such that the matrix entries vanish unless both source and target lie in $N_{R_{\mathrm{PL}}}(\partial\mathcal{K})$. For uniformly regular refinement families the same R_{PL} works for all members.

Proof outline of Theorem 2.8. The proof has four steps; the full version is in Section 6.19. (i) Self-adjointness and locality of $H_{\pm}^{\mathrm{PL}}(\mathcal{K})$ follow from the construction of the relative duality operator $S_{\mathcal{K}}^{\mathrm{PL}}$ (cap product with $[\mathcal{K}, \partial\mathcal{K}]$ followed by the canonical lift $j_{2-p}P_p^{\mathrm{rel}}$); the cap-product rule is local and the lift j is the standard quotient inclusion. (ii) The matrix-element comparison of P_D (closed double cap product) and $\widehat{P}$ (lifted relative cap product) is the content of Theorem 2.7: away from a fixed barycentric collar of $\partial\mathcal{K}$ the two cap-product blocks have identical matrix entries by the local nature of the simplicial cap-product formula. (iii) Consequently, the difference $R_{\partial,\pm} := H_{\pm}^{\partial}(\mathcal{K}) - H_{\pm}^{\mathrm{PL}}(\mathcal{K})$ has matrix entries supported in the collar $N_{R_{\mathrm{PL}}}(\partial\mathcal{K})$, where R_{PL} is the maximum propagation of the local cap-product rule plus the maximum propagation of the lift. (iv) The relative duality defect $B_{\mathcal{K}}S_{\mathcal{K}}^{\mathrm{PL}} + S_{\mathcal{K}}^{\mathrm{PL}}B_{\mathcal{K}}^{\dagger}$ is supported in the same collar by Theorem 2.6: the relative cap-product blocks satisfy $\partial_{2-p}^{\mathrm{rel}}P_p^{\mathrm{rel}} + (-1)^p P_{p+1}^{\mathrm{rel}}\delta_p = 0$ on the quotient $C_\bullet(\mathcal{K}, \partial\mathcal{K})$, so the lifted version has nonzero contributions only at the relative boundary, supported in the same fixed collar. For uniformly regular refinement families the local cap-product rule is the same, so the same R_{PL} works for all members.
Proof. See the online supplement, Section 6.19.

Lemma 2.9 (Spectral counts under collar-supported perturbations)**.** *If $R = P_W R P_W$ for a projection P_W of rank r, then $|N_{A+R}(t) - N_A(t)| \le r$ for all t and $|\mathrm{rank}\chi_I(A+R) - \mathrm{rank}\chi_I(A)| \le 2r$ for every interval I. Applied to $A = H_{\pm}^{\mathrm{PL}}(\mathcal{K})$ and $A + R = H_{\pm}^{\partial}(\mathcal{K})$, one may take $W = \ell^2(N_{R_{\mathrm{PL}}}(\partial\mathcal{K}))$. In particular, along refinement families with $r_n/d_n \to 0$, the normalized spectral distribution functions of the two boundary models share the same subsequential limits.*

Proof. See the online supplement, Section 6.21.

Proposition 2.10 (Fixed collars are negligible under quasi-uniform refinement)**.** *Let $(\mathcal{K}_n)_n$ be a uniformly regular refinement sequence of a fixed compact oriented PL surface, and suppose each $\mathcal{K}_n$ has full boundary subcomplex in the sense of Definition 2.1. Suppose moreover that $(\mathcal{K}_n)_n$ is a quasi-uniform shape-regular two-dimensional refinement with respect to a fixed PL metric: there is a mesh size $h_n \to 0$ such that boundary edge lengths are comparable to h_n, face areas are comparable to h_n^2, and the local incidence constants are uniformly bounded. For every fixed $R < \infty$, set*

$$d_n := \dim \mathcal{H}(\mathcal{K}_n), \qquad r_n(R) := \dim \ell^2(N_R(\partial\mathcal{K}_n)) .$$

Then

$$\frac{r_n(R)}{d_n} \longrightarrow 0.$$

In particular, for the fixed radius R_{PL} from Theorem 2.8, the boundary-collar dimension hypothesis in Theorem 3.16 holds for such refinement families.

Proof. See the online supplement, Section 6.22.

3 Finite-dimensional decoupling theory

The finite-dimensional analysis is perturbation theory specialized to the block structure produced by doubling and jamming. We use standard Schur-complement and spectral projection tools [16, 17, Chapter II], but keep the constants and separation hypotheses explicit because they are later checked on the computed matrices. The full proofs are collected in the online supplement, Section 1.

Lemma 3.1 (Schur complement resolvent formula). *Let* $T = \left(\begin{smallmatrix} A & C \\ C^\dagger & D \end{smallmatrix}\right)$ *be self-adjoint on* $\mathcal{H}_1 \oplus \mathcal{H}_2$ *and* $z \in \mathbb{C} \setminus \mathbb{R}$. *Then the* $(1,1)$ *block of* $(T-z)^{-1}$ *is* $\left(A - z - C(D-z)^{-1}C^\dagger\right)^{-1}$.

Write $D_M := D + M\mathbf{1}$ and $d_M(z) := \operatorname{dist}(z, \operatorname{spec}(D_M))$.

Theorem 3.2 (Compressed resolvent convergence with explicit $O(1/M)$ rate). *Let* $H_{\pm,M}$ *be as in* (2.1). *For any* $z \in \mathbb{C} \setminus \mathbb{R}$,

$$\left\| P_{\mathrm{orig}}(H_{\pm,M} - z)^{-1} P_{\mathrm{orig}} - (A-z)^{-1} \right\| \le \frac{\|C\|^2}{d_M(z)\,|\Im z|^2}. \tag{3.1}$$

In particular, if $M > \|D\| + |\Re z|$, $P_{\mathrm{orig}}(H_{\pm,M} - z)^{-1} P_{\mathrm{orig}} \to (A-z)^{-1}$ *in operator norm at rate* $O(1/M)$, *so the doubled-and-jammed Hamiltonian recovers the original-side boundary Hamiltonian* $A = H^{\partial}_{\pm}(\mathcal{K})$ *in the norm-resolvent sense as* $M \to \infty$.

Proof sketch; full proof in Supplement, Section 1. Write $R_M := P_{\mathrm{orig}}(H_{\pm,M} - z)^{-1} P_{\mathrm{orig}}$ and $R_0 := (A-z)^{-1}$. The Schur complement (Lemma 3.1) gives the resolvent identity $R_M - R_0 = R_M\, K_M\, R_0$ with the self-energy correction $K_M := C(D_M - z)^{-1}C^\dagger$. The resolvent norms satisfy $\|R_M\| \le |\Im z|^{-1}$ and $\|R_0\| \le |\Im z|^{-1}$, and the self-energy is bounded by $\|K_M\| \le \|C\|^2 / d_M(z)$. Multiplying gives (3.1). □

Corollary 3.3 (Boundary Hamiltonian as the large-jam limit). *Let* $H^{\partial}_{\pm}(\mathcal{K}) = A$ *be the original-side boundary Hamiltonian. For every* $z \in \mathbb{C} \setminus \mathbb{R}$, $P_{\mathrm{orig}}(H_{\pm,M} - z)^{-1} P_{\mathrm{orig}} \to (H^{\partial}_{\pm}(\mathcal{K}) - z)^{-1}$ *in operator norm with the explicit* $O(M^{-1})$ *bound* (3.1). *Thus the doubled-and-jammed Hamiltonian recovers a self-adjoint boundary Hamiltonian on the original side in the norm-resolvent sense; equivalently, by Theorem 2.8, the same limit is* $(H^{\mathrm{PL}}_{\pm}(\mathcal{K}) + R_{\partial,\pm} - z)^{-1}$.

Proof. See the online supplement, Section 6.26.

Corollary 3.4 (Degenerate full block-resolvent limit). *For* $z \in \mathbb{C}\setminus\mathbb{R}$, $(H_{\pm,M} - z)^{-1} \to (A-z)^{-1} \oplus 0$ *in operator norm at rate* $O(1/M)$, *with the explicit triangle inequality* $\|(H_{\pm,M} - z)^{-1} - ((A-z)^{-1} \oplus 0)\| \le \|C\|^2/(d_M(z)|\Im z|^2) + 2\|C\|/(d_M(z)|\Im z|) + 1/d_M(z) + \|C\|^2/(d_M(z)^2|\Im z|)$. *This is generalized norm-resolvent convergence in which the copy sector is sent to infinite energy.*

Proof. See the online supplement, Section 6.27.

Theorem 3.5 (First-order Schur-complement expansion). *Fix* $z \in \mathbb{C} \setminus \mathbb{R}$. *If* $M > \|D - z\|$ *and* $A^{(1)}_M := A - \frac{1}{M} CC^\dagger$, *then*

$$\left\| P_{\mathrm{orig}}(H_{\pm,M} - z)^{-1} P_{\mathrm{orig}} - (A^{(1)}_M - z)^{-1} \right\| \le \frac{\|C\|^2\, \|D - z\|}{M(M - \|D-z\|)\,|\Im z|^2} = O(M^{-2}). \tag{3.2}$$

Proof. See the online supplement, Section 6.29.

For a positively oriented simple closed contour $\Gamma \subset \rho(T)$ write the Riesz projection $P_\Gamma(T) := \frac{1}{2\pi i}\int_\Gamma (z - T)^{-1}\, dz$. Set $\delta_A(\Gamma) := \operatorname{dist}(\Gamma, \operatorname{spec}(A))$, $\delta_D(\Gamma) := \operatorname{dist}(\Gamma, \operatorname{spec}(D_M))$, and $\delta_H(\Gamma) := \delta_A(\Gamma) - \|C\|^2/\delta_D(\Gamma)$.

Theorem 3.6 (General contour-projection comparison). *Assume* $\delta_A(\Gamma) > 0$, $\delta_D(\Gamma) > 0$, $\delta_H(\Gamma) > 0$. *Then* $\Gamma \subset \rho(H_{\pm,M})$, $\operatorname{rank} P_\Gamma(H_{\pm,M}) = \operatorname{rank} P_\Gamma(A) + \operatorname{rank} P_\Gamma(D_M)$, *and*

$$\left\| P_\Gamma(H_{\pm,M}) - \left(P_\Gamma(A) \oplus P_\Gamma(D_M)\right) \right\| \le \tfrac{|\Gamma|}{2\pi}\, \eta_M(\Gamma) = O(M^{-1}), \tag{3.3}$$

where $\eta_M(\Gamma)$ is the explicit constant

$$\eta_M(\Gamma) = \frac{\|C\|^2}{\delta_A(\Gamma)\delta_H(\Gamma)\delta_D(\Gamma)} + \frac{2\,\|C\|}{\delta_H(\Gamma)\delta_D(\Gamma)} + \frac{\|C\|^2}{\delta_H(\Gamma)\delta_D(\Gamma)^2}.$$

Proof. See the online supplement, Section 6.30.

For $E_{\max} > 0$ define $g_M(E_{\max}) := \operatorname{dist}\big([-E_{\max}, E_{\max}], \operatorname{spec}(D_M)\big)$.

Theorem 3.7 (Bounded-energy spectral localization)**.** *If E is an eigenvalue of $H_{\pm,M}$ with $|E| \le E_{\max}$ and $g_M(E_{\max}) > 0$, then* $\operatorname{dist}(E, \operatorname{spec}(A)) \le \|C\|^2 / g_M(E_{\max})$.

Proof. See the online supplement, Section 6.31.

Corollary 3.8 (Spectral comparison with the relative boundary Hamiltonian)**.** *Under the hypotheses of Theorem 3.7,* $\operatorname{dist}\big(E, \operatorname{spec}(H_{\pm}^{\mathrm{PL}}(\mathcal{K}) + R_{\partial,\pm})\big) \le \|C\|^2 / g_M(E_{\max})$.

Proof. See the online supplement, Section 6.32.

Theorem 3.9 (Bounded-energy eigenvalues lie near the first-order effective operator)**.** *Let E be an eigenvalue of $H_{\pm,M}$ with $|E| \le E_{\max}$ and $g_M(E_{\max}) > 0$. Then*

$$\operatorname{dist}\big(E, \operatorname{spec}(A_M^{(1)})\big) \le \frac{\|C\|^2\,(\|D\| + E_{\max})}{M\, g_M(E_{\max})}. \tag{3.4}$$

In particular, if $M > \|D\| + E_{\max}$, then

$$\operatorname{dist}\big(E, \operatorname{spec}(A_M^{(1)})\big) \le \frac{\|C\|^2\,(\|D\| + E_{\max})}{M(M - \|D\| - E_{\max})}. \tag{3.5}$$

Proof. See the online supplement, Section 6.33.

Theorem 3.10 (Copy-side leakage bound)**.** *If (E, ψ) is an eigenpair of $H_{\pm,M}$ with $|E| \le E_{\max}$ and $g_M(E_{\max}) > 0$, the copy component satisfies $\|\psi_{\mathrm{copy}}\| \le \|C\|\ \|\psi_{\mathrm{orig}}\| / g_M(E_{\max}) = O(1/M)$.*

Proof. See the online supplement, Section 6.34.

Theorem 3.11 (Inertia reduction for large scalar jamming)**.** *Assume $M > \|D\|$, so $D_M = D + M\mathbf{1}$ is positive, and set $S_M(0) := A - CD_M^{-1}C^{\dagger}$. Then $N_-(H_{\pm,M}) = N_-(S_M(0))$, $\dim\ker H_{\pm,M} = \dim\ker S_M(0)$, and $\|S_M(0) - A\| \le \|C\|^2 / (M - \|D\|)$. Consequently, if $0 \notin \operatorname{spec}(A)$ and the perturbation satisfies $\|C\|^2 / (M - \|D\|) < \operatorname{dist}(0, \operatorname{spec}(A))$, then $H_{\pm,M}$ is invertible and $N_-(H_{\pm,M}) = N_-(A)$.*

Proof. See the online supplement, Section 6.35.

Corollary 3.12 (Monotonicity in the scalar jam scale)**.** *Let $M_2 > M_1$. The eigenvalues $\lambda_j(H_{\pm,M})$, listed in nondecreasing order with multiplicity, are nondecreasing in M. In particular, $M \mapsto N_-(H_{\pm,M})$ is nonincreasing.*

Proof. See the online supplement, Section 6.36.

Corollary 3.13 (No low-energy pure-copy states)**.** *Let* $E_{\max} > 0$ *and assume* $M > \|D\| + E_{\max}$*. The compression* $P_{\mathrm{orig}} Q_M : \mathrm{Ran} Q_M \to \mathcal{H}_{\mathrm{orig}}$ *is injective for* $Q_M = \chi_{[-E_{\max}, E_{\max}]}(H_{\pm,M})$*, so* $\mathrm{rank} Q_M \le \dim \mathcal{H}_{\mathrm{orig}}$*, and quantitatively* $\|P_{\mathrm{copy}}\psi\| \le (E_{\max} + \|C\|)/(M - \|D\|)\,\|\psi\|$ *for every* $\psi \in \mathrm{Ran} Q_M$*.*

Proof. See the online supplement, Section 6.37.

Proposition 3.14 (Inherited gap from the original-side block)**.** *Assume* $\mathrm{spec}(A) \cap (-g_A, g_A) = \emptyset$ *for some* $g_A > 0$ *and* $g_M(g_A) := \mathrm{dist}([-g_A, g_A], \mathrm{spec}(D_M)) > 0$*. With* $\Delta_M = \|C\|^2 / g_M(g_A)$*, if* $\Delta_M < g_A$ *then* $\mathrm{spec}(H_{\pm,M}) \cap (-g_A + \Delta_M, g_A - \Delta_M) = \emptyset$*.*

Proof. See the online supplement, Section 6.38.

Corollary 3.15 (Uniform direct-sum large-jam estimates)**.** *Under uniform bounds* $\bar{C} = \sup_n \|C_n\| < \infty$ *and* $\bar{D} = \sup_n \|D_n\| < \infty$*, the finite-M estimates of Theorems 3.2, 3.5, 3.7 and 3.10 hold uniformly in n for the direct-sum doubled-and-jammed family. The compressed resolvent error, the first-order compressed approximation, the spectral-localization distance, and the copy leakage carry the same explicit bounds with* $\|C\|$ *replaced by* $\bar{C}$ *and* $\|D\|$ *by* $\bar{D}$*.*

Proof. See the online supplement, Section 6.39.

Theorem 3.16 (Internal bulk spectral equivalence)**.** *Let* $(\mathcal{K}_n)_n$ *be finite triangulated oriented surfaces with full boundary subcomplex, with* $d_n = \dim \mathcal{H}(\mathcal{K}_n)$ *and uniform bounds on* $\|A_{\pm,n}\|, \|H^{\mathrm{PL}}_{\pm}(\mathcal{K}_n)\|, \|C_{\pm,n}\|, \|D_{\pm,n}\|$*. Let* $M_n \to \infty$ *and assume* $r_n/d_n \to 0$ *for the boundary-collar dimension* r_n *from Theorem 2.8. Then for every* $f \in C_0(\mathbb{R})$*,*

$$\frac{1}{d_n} \mathrm{Tr}\, f(H_{\pm,n,M_n}) - \frac{1}{d_n} \mathrm{Tr}\, f\big(H^{\mathrm{PL}}_{\pm}(\mathcal{K}_n)\big) \longrightarrow 0.$$

The intrinsic Poincaré–Lefschetz model, the doubled original-side compression, and the finite-M_n *jammed model have the same normalized bulk spectral subsequential limits. The statement uses no closed-bulk Chern theorem, no uniform doubled-bulk gap, and no interface-pairing hypothesis.*

Proof outline of Theorem 3.16. The proof combines three building blocks already established in this section; the full version is in Section 6.40. (i) The compressed-resolvent estimate Theorem 3.2 gives $\|P_{\mathrm{orig}}(H_{\pm,n,M_n} - z)^{-1} P_{\mathrm{orig}} - (A_n - z)^{-1}\| \to 0$ as $M_n \to \infty$ at the explicit $O(1/M_n)$ rate; in particular $\frac{1}{d_n}\big|\mathrm{Tr}\, f(H_{\pm,n,M_n}) - \mathrm{Tr}\, f(A_n)\big| \to 0$ for every $f \in C_0(\mathbb{R})$ by Stone–Weierstrass density of resolvents and the uniform bound $d_n^{-1}\,\mathrm{Tr}\,|\cdot| \le \|\cdot\|$. (ii) The collar-rank lemma Theorem 2.9, applied to $A_n = H^{\mathrm{PL}}_{\pm}(\mathcal{K}_n)$ and $A_n + R_{\partial,\pm,n} = H^{\partial}_{\pm}(\mathcal{K}_n)$, bounds the normalized trace difference by $O(r_n/d_n) \to 0$. (iii) Combining (i) with the identification $A_n = H^{\partial}_{\pm}(\mathcal{K}_n)$ and (ii) gives $\frac{1}{d_n}\big|\mathrm{Tr}\, f(H_{\pm,n,M_n}) - \mathrm{Tr}\, f(H^{\mathrm{PL}}_{\pm}(\mathcal{K}_n))\big| \to 0$ for every $f \in C_0(\mathbb{R})$. No closed-bulk Chern input or uniform doubled gap is needed; the hypothesis is only the uniform finite-dimensional operator-norm bound on the blocks $A_{\pm,n}, C_{\pm,n}, D_{\pm,n}$, and the negligible-collar condition $r_n/d_n \to 0$.
Proof. See the online supplement, Section 6.40.

The remaining estimates in this section – contour eigenvalue-count stability, monotonicity in the jam scale, no-copy low-energy bounds, an inherited-gap criterion, and the uniform direct-sum version used by the Roe construction – are stated together with their explicit constants and proved in the supplement, Section 1.

4 Roe-algebra identification of the doubled-and-jammed interface class

This section establishes the boundary-map identification used by the headline Roe-winding theorem. The closed-bulk gap and Chern value come from the closed universal-model theorem of Higson and Prodan [8, 9]. The Ewert–Meyer citation [29] fixes the coarse bulk–interface language and the Mayer–Vietoris pairing convention; for the doubled-and-jammed original/copy partition used in this paper, the identification of the Mayer–Vietoris boundary with the Roe connecting map is proved here.

4.1 Higson–Prodan universal closed-bulk input

Definition 4.1 (Universal Hamiltonian on a bounded-geometry triangulation)**.** Let K be a bounded-geometry closed oriented triangulated surface, let $\mathcal{H}(K) = \ell^2(K_0 \sqcup K_1 \sqcup K_2)$, let B be the simplicial boundary operator, and let S_K be the Higson–Prodan Hermitian Poincaré-duality operator constructed from cap product with the fundamental cycle. The two universal closed Hamiltonians are

$$H_\pm(K) := (B + B^\dagger) \pm S_K$$

on $\mathcal{H}(K)$.

Theorem 4.2 (Cited: Higson–Prodan closed-bulk gap and Chern)**.** (Higson–Prodan [8, 9]; statement as imported.) *Let $(K_n)_n$ be a sequence of finite closed oriented triangulated surfaces of uniformly bounded local geometry: there exist constants Δ, L, R_0 such that every vertex of K_n has simplicial-star valence at most Δ, every face has at most L edges in its star, and the link of every simplex is contained in a combinatorial ball of radius at most R_0; moreover (K_n) is a refinement sequence in the sense that each K_{n+1} admits a simplicial subdivision map to K_n. Let $H_\pm(K_n)$ be the universal Chern Hamiltonians of Theorem 4.1. Then:*

(i) (Uniform Fermi gap.) There exists $g_{\mathrm{HP}} > 0$, depending only on (Δ, L, R_0) and not on n, such that

$$\operatorname{spec}\big(H_\pm(K_n)\big) \cap (-g_{\mathrm{HP}}, g_{\mathrm{HP}}) = \varnothing \qquad \text{for all } n.$$

(ii) (Closed-bulk Chern integer.) Let $P_\pm(K_n) = \chi_{(-\infty,0)}(H_\pm(K_n))$ be the Fermi projection. The closed-bulk Bellissard–Connes/Higson–Prodan Chern pairing is

$$\mathrm{Ch}\big([P_\pm(K_n)]\big) = \pm 1,$$

with the sign determined by the chosen component H_+ versus H_- and by the orientation convention of Section 2 (Bloch torus orientation $dk_1 \wedge dk_2$ when the closed triangulation is periodic, and the canonical Higson–Prodan orientation otherwise).

This is the only external theorem cited anywhere in the proofs of this paper. Bibliographic data for the published PRL and the arXiv preprint are listed under [8] (Phys. Rev. Lett. ***136*** *(2026), no. 9, 096602, doi:10.1103/66x5-dvqq) and [9] (arXiv:2510.20862, doi:10.48550/arXiv.2510.20862).*

Lemma 4.3 (Doubled-jammed families satisfy HP hypotheses)**.** *Let $(\mathcal{K}_n)_n$ be a uniformly regular boundary refinement family of finite triangulated oriented surfaces with full boundary subcomplex (in the sense of Definition 2.1 together with uniform bounds on local valences and stars), and let $D(\mathcal{K}_n)$ be the simplicial doubles of Theorem 2.4. Then the closed sequence $(D(\mathcal{K}_n))_n$ satisfies the hypotheses of Theorem 4.2. Concretely:*

(i) *(Bounded geometry.) The vertex valences, face-star sizes, and link radii of $D(\mathcal{K}_n)$ are bounded uniformly in n by constants depending only on the corresponding bounds for $(\mathcal{K}_n)$ and on the doubling operation. Specifically, if every interior vertex of $\mathcal{K}_n$ has valence at most Δ_{int} and every boundary vertex has valence at most Δ_∂, then every vertex of $D(\mathcal{K}_n)$ has valence at most $\max(\Delta_{\mathrm{int}}, 2\Delta_\partial - 1)$; analogous bounds hold for face-star sizes and link radii by Theorem 2.4.*

(ii) *(Refinement structure.) If $(\mathcal{K}_n)$ is a refinement sequence (each $\mathcal{K}_{n+1}$ subdivides $\mathcal{K}_n$ and the subdivisions restrict to refinements of $\partial\mathcal{K}_n$), then $(D(\mathcal{K}_n))$ is a refinement sequence in the sense of Theorem 4.2: the doubled subdivision $D(\mathcal{K}_{n+1})$ admits a simplicial map to $D(\mathcal{K}_n)$ extending the subdivision on each side, gluing continuously across the shared boundary by the full-subcomplex condition of Definition 2.1.*

(iii) *(Closed oriented surface.) Each $D(\mathcal{K}_n)$ is a finite closed oriented triangulated surface (Theorem 2.4); the orientation on the original side is the orientation of $\mathcal{K}_n$, and on the copy side is the reversed orientation, consistent across $\partial\mathcal{K}_n$.*

Consequently Theorem 4.2 applies to $(D(\mathcal{K}_n))_n$: there is a uniform Fermi gap $g_{\mathrm{HP}} > 0$ for the closed doubled universal Hamiltonians $H_\pm(D(\mathcal{K}_n))$, and their closed-bulk Chern integers are $\mathrm{Ch}([P_\pm(D(\mathcal{K}_n))]) = \pm 1$.

Proof of Theorem 4.3. The valence, face-star, and link bounds are local combinatorial counts in the doubled complex $D(\mathcal{K}_n) = \mathcal{K}_n \cup_{\partial\mathcal{K}_n} \overline{\mathcal{K}_n}$; each interior simplex of $D(\mathcal{K}_n)$ is also an interior simplex of either $\mathcal{K}_n$ or $\overline{\mathcal{K}_n}$, inheriting its valence/star bounds, while each boundary simplex of $\mathcal{K}_n$ is shared with $\overline{\mathcal{K}_n}$ and has valence equal to the sum of valences from the two sides. The full-subcomplex condition Theorem 2.1 ensures that the gluing across $\partial\mathcal{K}_n$ does not create duplicate simplices, so $D(\mathcal{K}_n)$ is an abstract simplicial complex with these uniform bounds. The refinement property follows from the gluing-compatible subdivision of $\partial\mathcal{K}_n$ and the two-sided subdivision of $\mathcal{K}_n$ and $\overline{\mathcal{K}_n}$. That $D(\mathcal{K}_n)$ is a finite closed oriented triangulated surface is Theorem 2.4. □

Lemma 4.4 (Higson–Prodan closed-bulk gap, cited)**.** *For the universal Hamiltonians $H_\pm(K)$ on closed bounded-geometry triangulations covered by Theorem 4.2, there is a uniform Fermi gap $\mathrm{spec}(H_\pm(K)) \cap (-g_{\mathrm{HP}}, g_{\mathrm{HP}}) = \varnothing$ for some $g_{\mathrm{HP}} > 0$ independent of the refinement level. (Restatement of Theorem 4.2(i) for the closed-bulk gap input used throughout Section 4.)*

Lemma 4.5 (Higson–Prodan closed-bulk Chern number, cited)**.** *With $P_\pm(K) = \chi_{(-\infty,0)}(H_\pm(K))$, the Bellissard–Connes/Higson–Prodan closed-bulk Chern pairing satisfies $\mathrm{Ch}([P_\pm]) = \pm 1$, with the sign determined by the chosen component and by the orientation convention of Section 2. (Restatement of Theorem 4.2(ii) for the closed-bulk Chern input.)*

4.2 Roe extension and the interface boundary class

Let $X = \bigsqcup_n X(D(\mathcal{K}_n))$ be the coarse disjoint union of the doubled barycentric vertex sets, let X^+ be the original side, let X^- be the reflected-copy side, and let Y be a fixed coarse collar of the glued boundary. The choice of coarse collar radius does not affect the ideal $C^*(Y \subset X)$ (Proposition 5.5).

Lemma 4.6 (Roe algebra and closed-bulk Fermi projection)**.** *For X as above, the scalar Roe algebra $C^*(X)$ is unital. Under the Higson–Prodan gap of Lemma 4.4, the direct-sum Fermi projection $P_\pm = \chi_{(-\infty,0)}(\bigoplus_n H_\pm(D(\mathcal{K}_n)))$ belongs to $C^*(X)$.*

Lemma 4.7 (Coarse partition exact sequence)**.** *With $\mathcal{A} = C^*(X)$ and $\mathcal{I} = C^*(Y \subset X)$, the original/copy decomposition gives the quotient extension $0 \to \mathcal{I} \to \mathcal{A} \xrightarrow{\pi} \mathcal{A}/\mathcal{I} \to 0$, with boundary map $\partial_Y : K_0(\mathcal{A}/\mathcal{I}) \to K_1(\mathcal{I})$.*

Definition 4.8 (Interface projection and unitary)**.** Let $J^{\infty}_{\pm,M} = \bigoplus_n H_{\pm,M,n}$ be the doubled-and-jammed family and let $p_J := E_+\big(\pi(P_\pm)\big) \in \mathcal{A}/\mathcal{I}$. For any admissible cutoff h, $p_J = \pi(h(J^{\infty}_{\pm,M}))$, and the associated interface K_1-class is represented by $u_{J,h} := \exp(2\pi i\, h(J^{\infty}_{\pm,M})) \in \mathbf{1} + \mathcal{I}$, $[u_{J,h}] = \partial_Y([p_J])$.

4.3 Conventions for Theorem 4.9

The headline theorem fixes one set of conventions, used identically in the body proof and the appendix-level verifications:

(C1) *Representation.* The standard left-regular ℓ^2-representation of the scalar Roe algebra: $C^*(X) \subset \mathcal{B}(\ell^2(X))$ is the norm closure of locally compact finite-propagation operators acting on the canonical basis of unit point masses at each vertex of $X = \bigsqcup_n X(D(\mathcal{K}_n))$.

(C2) *Stabilization.* The scalar (not matrix-stabilized) Roe algebra. Standard Murray–von Neumann K-theoretic stabilization is invoked only where strictly necessary, namely in step (3) of the proof below.

(C3) *Sign convention for ∂.* The connecting map ∂_Y is the standard Higson–Roe boundary map in the six-term exact sequence of $0 \to \mathcal{I} \to \mathcal{A} \to \mathcal{A}/\mathcal{I} \to 0$, with the orientation choice that $\partial[\pi(P)] = [\exp(2\pi i\, \tilde{P})]$ for any self-adjoint lift $\tilde{P}$ of the quotient projection [10, 35].

(C4) *Side convention.* $X = X^+ \sqcup X^-$ is the original/copy decomposition; spectral cutoffs are taken on the negative real axis, $P_\pm = \chi_{(-\infty,0)}(H^{\infty}_{\pm})$.

(C5) *Quotient map.* $\pi : \mathcal{A} \to \mathcal{A}/\mathcal{I}$ is the canonical C^*-quotient by $\mathcal{I} = C^*(Y \subset X)$, with collar-radius independence.

The Ewert–Meyer Mayer–Vietoris boundary map for the same original/copy partition is denoted ∂_Y^{EM}; the internal compatibility theorem below identifies it with ∂_Y under (C1)–(C5).

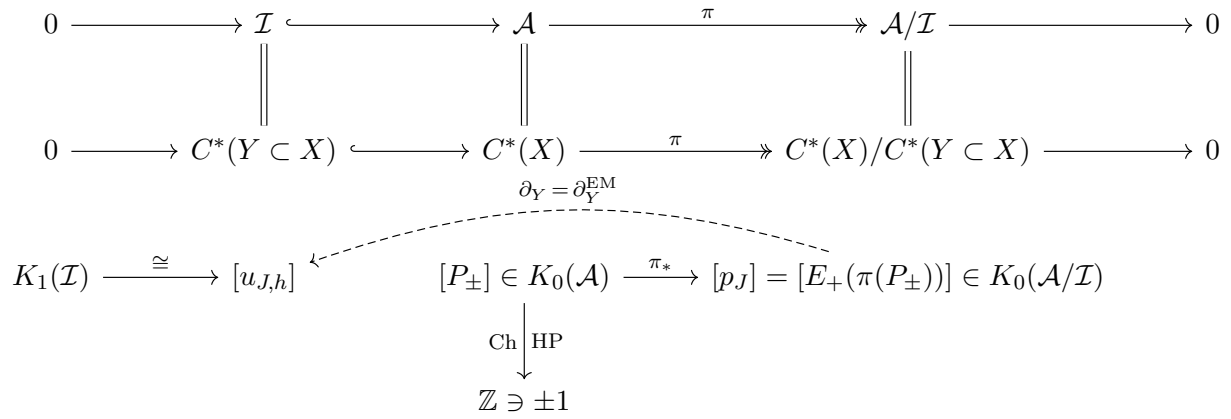


Figure 1: Commutative diagram for Theorem 4.9. The dashed arrow labelled $\partial_Y = \partial_Y^{\mathrm{EM}}$ records the internal compatibility theorem: the Roe connecting map and the Ewert–Meyer Mayer–Vietoris boundary coincide for the original/copy partition, sending $[p_J]$ to the interface K_1-class $[u_{J,h}] = [\exp(2\pi i\, h(J^{\infty}_{\pm,M}))]$. The downward arrow HP is Higson–Prodan's closed-bulk Chern character, which evaluates the bulk class to ± 1.

4.4 Internal compatibility theorem

Theorem 4.9 (Roe interface class from the Higson–Prodan closed bulk)**.** *Let $(\mathcal{K}_n)_n$ be a uniformly regular boundary refinement family whose closed doubles $D(\mathcal{K}_n)$ satisfy the hypotheses of Theorem 4.2 (verified for the doubled-jammed families of this paper in Theorem 4.3). Use the conventions (C1)–(C5). Let $\mathcal{A} = C^*(X)$, $\mathcal{I} = C^*(Y \subset X)$, and*

$$P_\pm = \chi_{(-\infty,0)}\left(\bigoplus_n H_\pm(D(\mathcal{K}_n))\right).$$

For every $M > B_{\mathrm{copy},-}$, the doubled-and-jammed family $J^\infty_{\pm,M}$ determines the quotient projection $p_J = E_+(\pi(P_\pm)) \in \mathcal{A}/\mathcal{I}$ and the K_1-class identity

$$\mathrm{Ind}_{\mathrm{int}}(J^\infty_{\pm,M}; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))) = [\exp(2\pi i\, h(J^\infty_{\pm,M}))] \in K_1(\mathcal{I})$$

for any admissible cutoff h. The Roe connecting map ∂_Y coincides on $[p_J]$ with the Ewert–Meyer Mayer–Vietoris boundary map ∂_Y^{EM} attached to the original/copy partition under (C1)–(C5); this identification is proved internally in Step (2) of the body proof and is the dashed arrow of Figure 1. The class $\partial_Y(E_+(\pi(P_\pm)))$ is independent of the admissible cutoff h and of the jam scale $M > B_{\mathrm{copy},-}$, and is preserved under refinement subsequences for which the Higson–Prodan Fermi gap is uniform.

Body proof of Theorem 4.9. **Roadmap.** The proof has four steps, in the order displayed by Figure 1. Steps (1), (2), and (4) are internal to this paper. Step (3) is the only step that uses an external theorem, namely Higson–Prodan [8, 9], packaged as Theorem 4.2. The convention block (C1)–(C5) fixes the representation, stabilization, sign, side, and quotient choices used uniformly across all four steps. Throughout the proof we work with one fixed short exact sequence of C^*-algebras

$$0 \longrightarrow \mathcal{I} \overset{\iota}{\hookrightarrow} \mathcal{A} \overset{\pi}{\twoheadrightarrow} \mathcal{A}/\mathcal{I} \longrightarrow 0, \qquad \mathcal{A} = C^*(X), \quad \mathcal{I} = C^*(Y \subset X), \tag{4.1}$$

to which we attach *two a priori distinct* six-term boundary maps: the Higson–Roe connecting map ∂_Y of Convention (C3), and the Ewert–Meyer Mayer–Vietoris boundary map ∂_Y^{EM} of [29]. The non-trivial content of Step (2) is the identity $\partial_Y = \partial_Y^{\mathrm{EM}}$ on $K_0(\mathcal{A}/\mathcal{I})$ for the original/copy partition.

Step (1): The Roe boundary map sends $[p_J]$ to the cutoff unitary class. By Lemma 4.6, $P_\pm \in \mathcal{A} = C^*(X)$ under the Higson–Prodan gap of Lemma 4.4 (which is Theorem 4.2(i)). So the quotient projection $p_J := E_+(\pi(P_\pm)) \in \mathcal{A}/\mathcal{I}$ is well defined as a self-adjoint idempotent in the quotient C^*-algebra $\mathcal{A}/\mathcal{I}$ of (4.1). The functional calculus of the quotient-sidewise statement (Proposition 5.28 in the supplement) provides, for any admissible cutoff h (continuous, equal to 1 on $[B_{\mathrm{copy},+}, \infty)$, to 0 on $(-\infty, -B_{\mathrm{copy},-}]$, and monotone in between), a self-adjoint element $h(J^\infty_{\pm,M}) \in \mathcal{A}$ with $\pi(h(J^\infty_{\pm,M})) = p_J$; see Definition 4.8 for the explicit lift. Since $h(J^\infty_{\pm,M})$ is a self-adjoint lift of an idempotent in the quotient, $(h(J^\infty_{\pm,M}))^2 - h(J^\infty_{\pm,M}) \in \mathcal{I}$, hence $\exp(2\pi i\, h(J^\infty_{\pm,M})) \in \mathbf{1} + \mathcal{I}$ and therefore defines a class in $K_1(\mathcal{I})$. Convention (C3) fixes the sign of the standard Higson–Roe K-theoretic connecting map for the short exact sequence (4.1), and the boundary-map lemma (Lemma 4.3) gives the labelled identity

$$\partial_Y([p_J]) = [\exp(2\pi i\, h(J^\infty_{\pm,M}))] \in K_1(\mathcal{I}), \tag{4.2}$$

independent of the admissible cutoff h. The class $[\exp(2\pi i\, h(J^\infty_{\pm,M}))]$ is the leftmost downstream class along the top row of Figure 1, and (4.2) is precisely the left dashed arrow. If the sign convention (C3) were replaced by its opposite (the alternate $\partial[\pi(P)] = -[\exp(2\pi i\, \tilde{P})]$ sometimes

seen in the literature), the right-hand side of (4.2) would acquire an overall minus sign, producing the inverse class in $K_1(\mathcal{I})$ and inverting all integer winding evaluations downstream; this is the auditable consequence of fixing the sign once and for all.

Step (2): The Roe and Ewert–Meyer boundary maps coincide for the original/copy partition. This is the key compatibility step. We claim that the Higson–Roe connecting map ∂_Y of Convention (C3) and the Ewert–Meyer Mayer–Vietoris boundary map ∂_Y^{EM} of [29] agree on $K_0(\mathcal{A}/\mathcal{I})$; concretely, viewed as homomorphisms $K_0(\mathcal{A}/\mathcal{I}) \to K_1(\mathcal{I})$, they are the *same* homomorphism.

(2a) Both maps come from one and the same short exact sequence. The Roe-extension map ∂_Y is the boundary map of the six-term exact sequence attached to (4.1) with the three named ingredients $(\mathcal{A}, \mathcal{I}, \pi) = (C^*(X), C^*(Y \subset X), \pi_{\mathrm{Roe}})$. The Ewert–Meyer Mayer–Vietoris boundary ∂_Y^{EM} is the boundary map of the six-term exact sequence attached to the bulk–interface pair $(X^+, X^-; Y)$ used in [29]; for the doubled-and-jammed setting at hand, the constituent algebras are: the bulk algebra is $C^*(X)$ for the same $X = X^+ \sqcup X^-$; the interface ideal is $C^*(Y \subset X)$ for the same Y; and the quotient is the canonical C^*-quotient $C^*(X)/C^*(Y \subset X)$, with the same C^*-quotient map. The coarse-excision statement (Proposition 5.25) is exactly the assertion that these triples match: the bulk algebra $\mathcal{A}_{\mathrm{EM}}$ of [29] is $C^*(X)$, the interface ideal $\mathcal{I}_{\mathrm{EM}}$ is $C^*(Y \subset X)$, and the EM quotient $\mathcal{A}_{\mathrm{EM}}/\mathcal{I}_{\mathrm{EM}}$ is the canonical C^*-quotient. We record this comparison as the labelled identity

$$(\mathcal{A}, \mathcal{I}, \mathcal{A}/\mathcal{I}, \pi)_{\mathrm{Roe}} \;=\; (\mathcal{A}, \mathcal{I}, \mathcal{A}/\mathcal{I}, \pi)_{\mathrm{EM}}. \tag{4.3}$$

(2b) Both maps are induced by the same lift. For the Roe extension, the boundary class $\partial_Y([\pi(P)])$ is by definition (Convention (C3)) the class $[\exp(2\pi i\, \tilde{P})]$ for any self-adjoint lift $\tilde{P}$ of $\pi(P)$; see [10, §6] and [35]. For the Ewert–Meyer extension, the Mayer–Vietoris boundary on a coarsely excisive partition is the connecting map of the same six-term exact sequence, with the same lift recipe: a self-adjoint operator on $\ell^2(X)$ of finite propagation that maps to $\pi(P)$ under the quotient is the EM lift; see [29, §3] where the Mayer–Vietoris boundary is constructed as the boundary map of the underlying C^*-extension. Since the only datum required to define either boundary map on a single element of $K_0(\mathcal{A}/\mathcal{I})$ is the lift and the sign convention, and both are fixed identically by (4.1) and Convention (C3), we record the labelled identity

$$\partial_Y([\pi(P)]) \;=\; [\exp(2\pi i\, \tilde{P})] \;=\; \partial_Y^{\mathrm{EM}}([\pi(P)]) \qquad \text{for every self-adjoint lift } \tilde{P} \text{ of } \pi(P). \tag{4.4}$$

By naturality of the six-term exact sequence (a standard property of any short exact sequence of C^*-algebras; [19, §8]), two boundary maps for the same short exact sequence with the same lift and the same sign convention coincide as homomorphisms on $K_0(\mathcal{A}/\mathcal{I})$: there is no degree of freedom remaining.

(2c) Side and quotient consistency. Convention (C4) fixes $X = X^+ \sqcup X^-$ as the original/copy partition; the spectral cutoff is on $(-\infty, 0)$, so $P_\pm = \chi_{(-\infty,0)}(H_\pm^\infty)$ is the negative Fermi projection. This is the same side and spectral choice used by [29] for the negative-energy Fermi-projection class; the alternative $\chi_{(0,\infty)}$ cutoff would produce $\mathbf{1} - P_\pm$ instead, and would flip the sign of every quotient class downstream. The quotient projection $p_J = E_+(\pi(P_\pm))$ is computed by the quotient-sidewise identity (Proposition 5.28); the splitting identity $\mathcal{A}/\mathcal{I} = E_+(\mathcal{A}/\mathcal{I}) \oplus E_-(\mathcal{A}/\mathcal{I})$ (Proposition 5.27) reduces the comparison to the positive side because the negative-side comparison operator has trivial Fermi projection by the trivial-jam statement (Proposition 5.23). If the side convention (C4) were replaced by the opposite partition (copy on the original side), the labelled identity (4.4) would still hold but with the roles of (X^+, X^-) interchanged; the trivial-jam statement and the splitting

identity guarantee that the answer does not depend on which side is chosen as X^+, only that the splitting itself is fixed.

(2d) Quotient convention. Both boundary maps use the same C^*-quotient $\pi : \mathcal{A} \to \mathcal{A}/\mathcal{I}$ (Convention (C5)), with collar-radius independence (Proposition 5.5). An alternative quotient by a smaller or larger ideal would change $\mathcal{A}/\mathcal{I}$ and hence both boundary maps; the equality $\partial_Y = \partial_Y^{\mathrm{EM}}$ is a statement about the *specific* quotient by $C^*(Y \subset X)$, as fixed in (4.1), and not about boundary maps for unrelated ideals.

Combining (4.3), (4.4), and the side/quotient checks (2c)–(2d), we conclude the labelled compatibility identity

$$\partial_Y \;=\; \partial_Y^{\mathrm{EM}} \qquad \text{on} \qquad K_0(\mathcal{A}/\mathcal{I}), \tag{4.5}$$

which is the dashed arrow of Figure 1 labelled $\partial_Y = \partial_Y^{\mathrm{EM}}$. In particular, $\partial_Y^{\mathrm{EM}}([p_J]) = \partial_Y([p_J]) = [\exp(2\pi i\, h(J^\infty_{\pm,M}))]$ by composing (4.2) with (4.5). We emphasise that this identity uses no part of the closed-bulk Chern statement; only the gap of Theorem 4.2(i) is invoked, via Lemma 4.6, to ensure $P_\pm \in \mathcal{A}$. If either the EM lift recipe or the EM sign convention differed from (C3), the right-hand side of (4.4) could pick up a sign, producing the inverse class in $K_1(\mathcal{I})$ and breaking the labelled equality (4.5); both references [29, 10] use the same six-term-exact-sequence sign and lift recipe (the difference between the two reduces to a choice of book-keeping for the Mayer–Vietoris pairing on coarsely excisive partitions), so the convention agreement is verified.

Step (3): Closed-bulk Chern from Higson–Prodan. By Lemma 4.5, which restates Theorem 4.2(ii) for closed bounded-geometry triangulated surfaces, the closed-bulk Fermi projection has Chern number $\mathrm{Ch}([P_\pm]) = \pm 1$. This is the only step that uses an external theorem, namely Theorem 4.2; the doubled families satisfy the HP hypotheses by Theorem 4.3. The integer evaluation of the K_1-class identity from Step (2) is taken up by the two corollaries (Theorem 4.10 for the internal Toeplitz periodic case; Theorem 4.11 for the general conditional case). This is the downward Chern-character arrow of Figure 1 labelled HP. If the orientation convention of Section 2 were flipped (e.g. $dk_2 \wedge dk_1$ instead of $dk_1 \wedge dk_2$ on the Bloch torus), the closed-bulk Chern sign would flip and the integer winding downstream would change sign correspondingly; the sign-ledger appendix (Section 6.1) records the explicit chain.

Step (4): Finite-M quotient identification and K_1-limit. For every $M > B_{\mathrm{copy},-}$, the trivial-jam statement (Proposition 5.23) gives strict positivity of the negative comparison operator, so its negative Fermi projection vanishes. The positive comparison operator is $T^+ = H^\infty_\pm$ with negative Fermi projection $P_\pm$, and the quotient-sidewise identification (Proposition 5.28) gives $p_J = E_+(\pi(P_\pm)) + E_-(\pi(0)) = E_+(\pi(P_\pm))$ at every finite jam scale $M > B_{\mathrm{copy},-}$. The corresponding K_1-class is independent of M (Corollary 6.38), so the limit $M \to \infty$ is well defined as a single $K_1(\mathcal{I})$-class. Composing the four steps—namely (4.2) from Step (1), (4.5) from Step (2), Theorem 4.2(ii) from Step (3), and the quotient-sidewise identification of this Step (4)—gives the claimed K_1-class identity $\mathrm{Ind}_{\mathrm{int}}(J^\infty_{\pm,M}; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))) = [\exp(2\pi i\, h(J^\infty_{\pm,M}))]$. This is the full top row of Figure 1 traversed from $[P_\pm]$ on the right, through π_* to $[p_J]$, and through the dashed $\partial_Y = \partial_Y^{\mathrm{EM}}$ arrow back to $[u_{J,h}]$ on the left. Detailed computational verifications of the cutoff-functional-calculus identity, the norm-continuous homotopy along admissible cutoffs, and the explicit lift $h(J^\infty_{\pm,M})$ are collected in the supplement, Section 6.119. If the quotient-sidewise identification of (4) used a different (e.g. Schur-complement) lift, the K_1-class $[\exp(2\pi i\, h(J^\infty_{\pm,M}))]$ would still equal $\partial_Y([p_J])$ by (4.2), but the explicit lift formula on the right-hand side would differ; the cutoff-independence of the class (Corollary 6.38) is what makes the identity robust under the choice of admissible h. □

Corollary 4.10 (Integer winding in the Toeplitz periodic case)**.** *Suppose $Y \subset X$ is the straight periodic interface of the periodic triangular class of Section 5, with primitive tangential/positive-normal basis (t, n) and $\varepsilon(t, n) = \det[t\ n] \in \{\pm 1\}$. Then the K_1-class $\partial_Y(E_+(\pi(P_\pm)))$ of Theorem 4.9 maps under the Toeplitz extension to the integer*

$$\mathrm{wind}\big(\partial_Y(E_+(\pi(P_\pm)))\big) = \varepsilon(t,n)\ \mathrm{Ch}\big(E_-(H_\pm^\triangle)\big) = \mp\,\varepsilon(t,n) = \mp\det[t\ n] \in \{\pm 1\},$$

identifying the K_1-class with $\mathbb{Z}$ via the Toeplitz quotient in Theorem 5.7. The two displayed equalities use, in order, the Toeplitz pairing formula $\mathrm{wind}(\partial_{\mathrm{T},t,n}[P]) = \varepsilon(t,n)\ \mathrm{Ch}(P)$ of Corollary 5.8 and the Chern signs $\mathrm{Ch}(E_-(H_+^\triangle)) = -1$, $\mathrm{Ch}(E_-(H_-^\triangle)) = +1$ of Theorem 5.5; the resulting upper/lower sign reading agrees with the straight-edge formula $\mathrm{wind} = -\det[t\ n]$ for $H_+^\triangle$ and $\mathrm{wind} = +\det[t\ n]$ for $H_-^\triangle$ of Corollary 5.9. The integer-pairing statement in this case is fully internal: the closed-bulk gap and Chern signs are proved in Theorem 5.2 and Theorem 5.5, and the Toeplitz identification of K_1 with $\mathbb{Z}$ is the standard BDF–Toeplitz extension recalled in Theorem 5.7 and Corollary 5.8.

Proof outline of Theorem 4.10. The K_1-class identity $\partial_Y(E_+(\pi(P_\pm))) = [\exp(2\pi i\, h(J^\infty_{\pm,M}))]$ of Theorem 4.9 is the input. Compressing the positive-normal direction n of the periodic interface to a half-space realizes the interface ideal $\mathcal{I} = C^*(Y \subset X)$ as the kernel of the Toeplitz extension $0 \to C(S^1, \mathcal{K}) \to \mathcal{T}_{\mathrm{per}} \to C(\mathbb{T}^2) \to 0$ of Theorem 5.7, whose half-space-direction connecting map $\partial_{\mathrm{T},t,n}$ satisfies $\mathrm{wind}(\partial_{\mathrm{T},t,n}[P]) = \varepsilon(t,n)\ \mathrm{Ch}(P)$ by Theorem 5.8. With $P = E_-(H_\pm^\triangle)$ and the Chern signs of Theorem 5.5, the displayed equality holds: the trivial-jam statement gives $[P_-] = [0]$, so only the positive-side contribution remains. Full verification of the cutoff-functional-calculus identification of the K_1-class with the Toeplitz winding integer is in Section 6.120.

Proof. See the online supplement, Section 6.120.

Corollary 4.11 (Conditional integer winding for general partitions)**.** *Suppose that the K_1-class of the interface ideal $\mathcal{I} = C^*(Y \subset X)$ admits an integer-valued pairing $\langle \cdot, \cdot \rangle : K_0(\mathcal{A}/\mathcal{I}) \times K_1(\mathcal{I}) \to \mathbb{Z}$ of Chern-jump type, in the sense that $\langle [p_J], \partial_Y([p_J]) \rangle = \mathrm{Ch}([P_+]) - \mathrm{Ch}([P_-])$ whenever both sides are defined and ∂_Y is the Higson–Roe connecting map of Convention (C3). Then the pairing of $\partial_Y(E_+(\pi(P_\pm)))$ with $[p_J]$ is ± 1. The Toeplitz periodic case (Corollary 4.10) is the special case where the integer pairing is internal; the general case is conditional on the existence of an Ewert–Meyer/Chern-jump compatible integer pairing whose construction outside the periodic class is not carried out in this paper.*

Proof. See the online supplement, Section 6.121.

Corollary 4.12 (Finite-M cutoff stability)**.** *For every $M > B_{\mathrm{copy},-}$, the $K_1(\mathcal{I})$-class $\partial_Y(E_+(\pi(P_\pm)))$ of Theorem 4.9 is independent of the jam scale M: equality of classes in $K_1(\mathcal{I})$ follows from the cutoff homotopy and jam-scale-independence statements (Corollary 6.38 and Proposition 5.28).*

4.5 Stability and cross-checks

Lemma 4.13 (Collar-supported stability without norm-smallness)**.** *Let V be any bounded self-adjoint perturbation in $C^*(Y \subset X)$, with no small-norm assumption, and suppose the positive and negative comparison gaps used to define the sidewise interface remain open. Then the quotient projection and K_1-class $\partial_Y(E_+(\pi(P_\pm)))$ of Theorem 4.9 are unchanged. In the straight periodic Toeplitz case (Corollary 4.10), the resulting integer winding is also unchanged.*

Proof. See the online supplement, Section 6.123.

Lemma 4.14 (Bounded-geometry stability)**.** *Along a norm-continuous bounded-geometry deformation of the closed doubled universal Hamiltonians that preserves the Higson–Prodan Fermi gap, the interface K_1-class of Theorem 4.9 is constant.*

Proof. See the online supplement, Section 6.124.

Lemma 4.15 (Refinement stability)**.** *For a refinement family $K_n \to K_\infty$ in bounded geometry, if the Higson–Prodan theorem gives a uniform gap g_{HP} and the jam scales satisfy $M_n > B_{\mathrm{copy},-} + 1$, $M_n \to \infty$, then the quotient class and K_1-class of Theorem 4.9 are eventually constant under subsequences or cofinal refinements.*

Proof. See the online supplement, Section 6.125.

Remark 4.16 (Periodic triangular cross-check). For periodic triangular doubles, Theorem 4.9 agrees with the internal Bloch gap, Chern, and straight Toeplitz winding calculations of Section 5; details in the supplement, Section 6.126.

Remark 4.17 (Finite-defect cross-check). For doubled finite-defect families whose defects remain in a fixed collar or in finitely many coarse components, Theorem 4.9 gives the same interface class as the finite-defect robustness statements in the supplement, Section 6.5; details in Section 6.127.

Remark 4.18 (Hexagonal and square cross-check). For the supplement's hexagonal and square symbols (Section 6.3), whenever their closed-bulk class is identified with the corresponding Higson–Prodan closed-bulk class or with the internally computed periodic Chern class through a gapped homotopy, Theorem 4.9 returns the same interface K_1-class, and the Toeplitz integer winding agrees with Corollary 4.10 in the periodic case; details in Section 6.128.

Remark 4.19 (Tail-split and annular cross-check). For the tail-split and annular families of the supplement (Section 6.7), the boundary class of Theorem 4.9 restricts componentwise to the same Toeplitz quotient coordinates used there; details in Section 6.129.

Remark 4.20 (Input synthesis). A naive accounting separates the closed doubled-bulk gap, the closed-bulk Chern pairing, and the Ewert–Meyer comparison into three distinct inputs. Theorem 4.9 packages the first two as the single Higson–Prodan closed theorem and proves the third for the doubled-jammed original/copy partition. The only cited closed-model theorem needed for the K_1-class identity is the Higson–Prodan theorem. The integer winding evaluation is internal in the periodic triangular Toeplitz case (Theorem 4.10) and is recorded as a conditional corollary for general partitions (Theorem 4.11).

The companion coarse interface class construction, which places the doubled-and-jammed refinement family in a Roe-algebra construction producing the same quotient projection and K_1-class from uniform locality, boundedness, and an available closed-bulk gap, is recorded in the supplement, Section 5. Its main statement (Theorem 5.35) and the Ewert–Meyer-language restatement (Theorem 5.38) consolidate the same conclusion.

5 A periodic triangular bulk input

One closed-bulk input is checked directly here, rather than imported from Higson–Prodan. Let $z_j = e^{ik_j}$, $k = (k_1, k_2) \in \mathbb{T}^2$, and consider the $\mathbb{Z}^2$-periodic triangular triangulation with one vertex v, three oriented edges a, b, c, and two oriented faces f, g per fundamental cell. The Bloch fiber is

$$\mathcal{H}(k) = \mathbb{C}\langle v\rangle \oplus \mathbb{C}\langle a, b, c\rangle \oplus \mathbb{C}\langle f, g\rangle.$$

With respect to this ordered basis, the boundary blocks are

$$\partial_1(k) = \begin{bmatrix} z_1 - 1 & z_2 - 1 & z_1 z_2 - 1 \end{bmatrix}, \qquad \partial_2(k) = \begin{bmatrix} 1 & -z_2 \\ z_1 & -1 \\ -1 & 1 \end{bmatrix}.$$

A direct multiplication gives $\partial_1(k)\partial_2(k) = 0$. The cap-product blocks are

$$P_0(k) = \begin{bmatrix} z_1^{-1} z_2^{-1} \\ z_2^{-1} \end{bmatrix}, \qquad P_1(k) = \begin{bmatrix} 0 & z_1^{-1} & 0 \\ 0 & 0 & 0 \\ -z_2^{-1} & 0 & 0 \end{bmatrix}, \qquad P_2(k) = \begin{bmatrix} 1 & 1 \end{bmatrix},$$

and the Hermitian duality block is

$$S_0(k) = \tfrac{i}{2}(P_2(k)^* + P_0(k)), \quad S_1(k) = \tfrac{i}{2}(P_1(k)^* - P_1(k)), \quad S_2(k) = -\tfrac{i}{2}(P_0(k)^* + P_2(k)),$$

so that $H_{\pm}^{\triangle}(k) = B(k) + B(k)^* \pm S(k)$ is Hermitian.

Proposition 5.1 (Periodic triangular characteristic polynomial)**.** *Set*

$$F(k) = 28 - 8\cos k_1 - 6\cos k_2 - 6\cos(k_1 + k_2), \qquad G(k) = \sin k_2 + \sin(k_1 + k_2).$$

Then

$$\det\Big(\lambda \mathbf{1} - H_{\pm}^{\triangle}(k)\Big) = \big(\lambda^2 - \tfrac{F(k)}{4}\big)\big(\lambda^4 - \tfrac{13}{2}\lambda^2 + \tfrac{F(k)}{4} \pm \lambda G(k)\big),$$

with one exact pair of eigenvalue branches $\lambda = \pm\frac{1}{2}\sqrt{F(k)}$.

Proof. See the online supplement, Section 6.41.

Theorem 5.2 (Periodic triangular bulk gap)**.** *For every* $k \in \mathbb{T}^2$, $H_{\pm}^{\triangle}(k)$ *is invertible. More precisely,*

$$\det H_{\pm}^{\triangle}(k) = -\tfrac{1}{16}\left(-28 + 8\cos k_1 + 6\cos k_2 + 6\cos(k_1 + k_2)\right)^2 \le -4.$$

Moreover $\|H_{\pm}^{\triangle}(k)\| \le 8$, *hence* $\operatorname{dist}\big(0, \operatorname{spec} H_{\pm}^{\triangle}(k)\big) \ge 2^{-13}$ *for all* $k \in \mathbb{T}^2$, *and the negative Bloch bundle has rank* 3.

Proof. See the online supplement, Section 6.42.

Corollary 5.3 (Small periodic perturbations preserve the triangular gap)**.** *If* $V(k)$ *is a continuous self-adjoint* 6×6 *matrix over* $\mathbb{T}^2$ *with* $\sup_k \|V(k)\| < 2^{-13}$, *then* $H_{\pm}^{\triangle}(k) + V(k)$ *is invertible for every* $k \in \mathbb{T}^2$, *and its negative Bloch bundle has rank* 3.

Proof. See the online supplement, Section 6.43.

Proposition 5.4 (Positive duality strength homotopy)**.** *For* $m > 0$, *the operator* $H_{\pm,m}^{\triangle}(k) = B(k) + B(k)^* \pm mS(k)$ *is invertible for every* $k \in \mathbb{T}^2$; *explicitly,*

$$\det H_{\pm,m}^{\triangle}(k) = -\tfrac{1}{16}\big(8(3 - c_1 - c_2 - c_{12}) + 2m^2(2 + c_2 + c_{12})\big)^2,$$

with $c_j = \cos k_j$, $c_{12} = \cos(k_1 + k_2)$. *All positive values of the duality strength are connected by a zero-gapped norm-continuous homotopy.*

Proof. See the online supplement, Section 6.44.

Theorem 5.5 (Periodic triangular Chern signs)**.** *For the periodic triangular Bloch Hamiltonians above, the rank-three negative Bloch bundles have*

$$\mathrm{Ch}\big(E_-(H_+^{\triangle})\big) = -1, \qquad \mathrm{Ch}\big(E_-(H_-^{\triangle})\big) = +1.$$

Proof. See the online supplement, Section 6.45.

Corollary 5.6 (Periodic Chern stability)**.** *With* $g_{\pm,m} := \min_k \operatorname{dist}(0, \operatorname{spec} H_{\pm,m}^{\triangle}(k)) > 0$*, every continuous self-adjoint periodic perturbation* $V(k)$ *with* $\sup_k \|V(k)\| < g_{\pm,m}$ *keeps* $H_{\pm,m}^{\triangle}(k) + V(k)$ *gapped at zero and preserves the Chern number. The same holds for* $H_{\pm}^{\triangle}$ *when* $\sup_k \|V(k)\| < 2^{-13}$*. In particular, sufficiently small finite-range self-adjoint periodic perturbations of the triangular model do not change the Chern signs.*

Proof. See the online supplement, Section 6.46.

As a numerical check, the companion script applies the gauge-invariant lattice formula of Fukui–Hatsugai–Suzuki [7] to the rank-three negative spectral projection, checking the returned integer, the link denominators, and the plaquette branch margins; the computation records the same signs as Theorem 5.5. The antiunitary equivalence $H_-^{\triangle}(k) = \overline{H_+^{\triangle}(-k)}$ explains the opposite signs through complex conjugation of the Bloch bundle, with the base map $k \mapsto -k$ orientation-preserving in two dimensions.

Theorem 5.7 (Toeplitz evaluation for a straight periodic interface)**.** *Let* $H(k_1, k_2)$ *be a continuous finite-range self-adjoint* $N \times N$ *Bloch Hamiltonian on* $\mathbb{T}^2$*, gapped at zero, and let* $P(k_1, k_2) = \chi_{(-\infty,0)}(H(k_1, k_2))$*. Write* $\mathrm{Ch}(P) = \frac{1}{2\pi i}\int_{\mathbb{T}^2} \mathrm{tr}(P\, dP \wedge dP) \in \mathbb{Z}$ *with the orientation* $dk_1 \wedge dk_2$*. Compress the second lattice direction to a half-space. After Fourier transform in the first direction, the straight-edge extension is the Toeplitz extension* $0 \to C(S^1, \mathcal{K}) \to \mathcal{T}_{\mathrm{per}} \xrightarrow{\sigma} C(\mathbb{T}^2) \to 0$ *with connecting map* $\partial_{\mathrm{T}} : K_0(C(\mathbb{T}^2)) \to K_1(C(S^1))$ *satisfying* $\mathrm{wind}(\partial_{\mathrm{T}}([P])) = \mathrm{Ch}(P)$*. The sign is fixed by the above orientation and by taking the compressed coordinate to be the positive normal direction; see, for example, [20, 19, 21, 30].*

Proof. See the online supplement, Section 6.52.

Corollary 5.8 (Primitive-direction straight periodic interfaces)**.** *Let* $t, n \in \mathbb{Z}^2$ *be a lattice basis with* t *the tangential direction of a straight periodic interface and* n *the compressed positive normal direction. With* $\varepsilon(t, n) := \det[t\ n] \in \{\pm 1\}$*, the half-space Toeplitz boundary map in the* η*-direction* $(k = \kappa t + \eta n)$ *satisfies* $\mathrm{wind}(\partial_{\mathrm{T},t,n}([P])) = \varepsilon(t, n)\ \mathrm{Ch}(P)$*. In particular, every primitive straight periodic edge direction is covered by a lattice change of variables.*

Proof. See the online supplement, Section 6.53.

Corollary 5.9 (Periodic straight-edge winding from Toeplitz pairing)**.** *For the periodic triangular bulk models* $H_{\pm}^{\triangle}$*, every primitive straight periodic interface with tangential/positive-normal basis* (t, n) *has interface winding*

$$\mathrm{wind} = -\det[t\ n] \quad \text{for } H_+^{\triangle}, \qquad \mathrm{wind} = \det[t\ n] \quad \text{for } H_-^{\triangle}.$$

The same formula holds for every fixed positive-duality-strength triangular model and for every small finite-range periodic perturbation covered by Corollary 5.6. Finite-cover, finite-defect, finite-corner, split, and tail-split variants are the appendix statements collected in the supplement, Section 6.4.

Proof. See the online supplement, Section 6.60.

Hexagonal and square two-band benchmarks are recorded in the supplement, Section 6.3. Sign-ledger and orientation conventions are recorded in the supplement, Section 6.1.

6 Numerical checks

The numerical material is collected in the online supplement, Section 7, with reproducibility details in Sections 7.1 and 7.2. The computations are not proof inputs; they document finite-matrix behavior for the mechanisms proved above. We retain two numerical figures in the main paper that bear directly on the headline results: the intrinsic Poincaré–Lefschetz comparison (Figure 2) and the finite-M decoupling log-log scaling (Figure 3).

Intrinsic boundary comparison. Figure 2 compares the intrinsic Poincaré–Lefschetz boundary Hamiltonian $H_+^{\mathrm{PL}}(\mathcal{K})$, the doubled original-side compression $H_+^{\partial}(\mathcal{K})$, and the finite-$M$ jammed model on a disk-like sample. The collar correction $R_{\partial,+} = H_+^{\partial}(\mathcal{K}) - H_+^{\mathrm{PL}}(\mathcal{K})$ is not norm-small, but it is confined to the boundary collar predicted by Theorem 2.8, and the jammed low-energy spectrum tracks the compressed boundary model in line with the Schur-complement bounds of Theorem 3.2 and Theorem 3.5. The numerical refinement scaling of $R_{\partial,+}$ – recorded in the supplement – is consistent with the boundary-rank interpretation in Lemma 2.9.

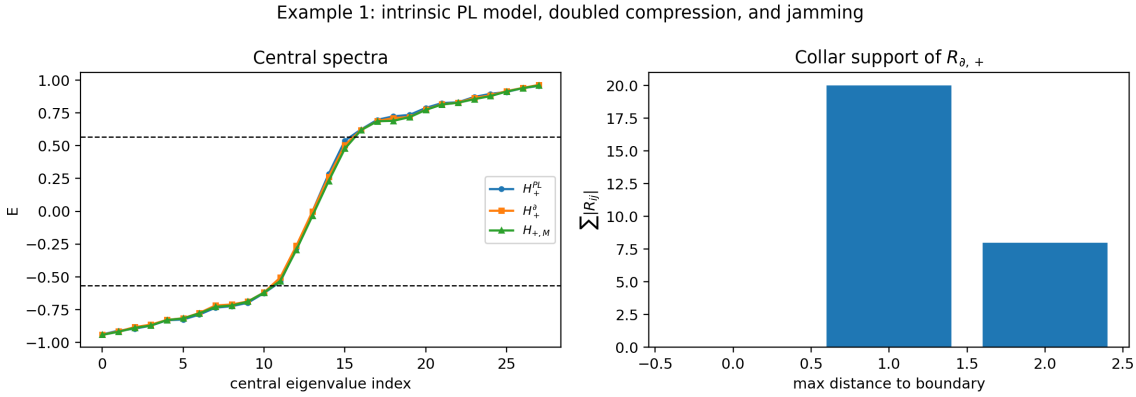
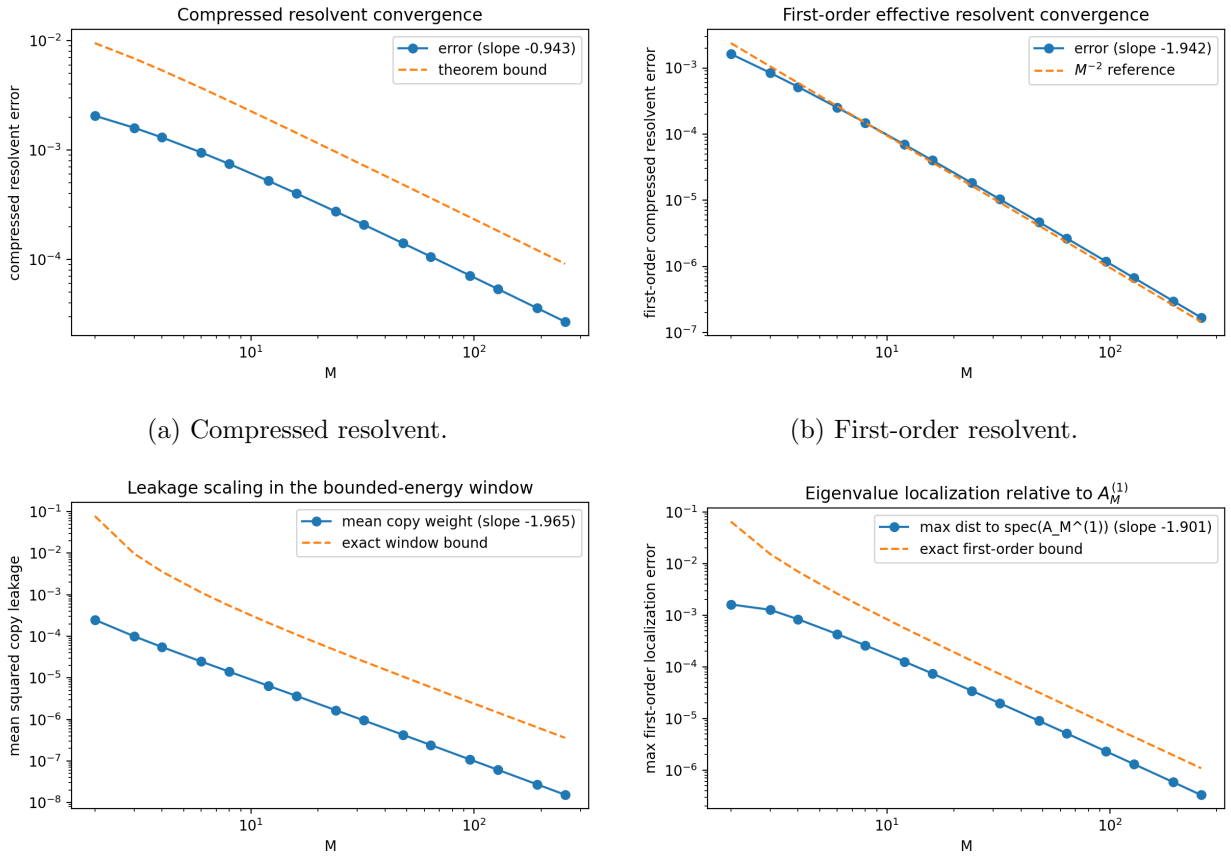


Figure 2: Central spectra of $H_+^{\mathrm{PL}}(\mathcal{K})$, $H_+^{\partial}(\mathcal{K})$, and $H_{+,M}$ on the disk-like example, together with the measured support distribution of the collar correction $R_{\partial,+}$. The correction is not norm-small, but it is boundary-collar supported, as predicted by Theorem 2.8.

Finite-M log-log scalings. Figure 3 reports the finite-M rate diagnostics. The compressed resolvent error follows the M^{-1} bound of Theorem 3.2; the first-order resolvent correction, squared copy leakage, and first-order eigenvalue localization follow the M^{-2} predictions of Theorems 3.5 and 3.10 and the first-order eigenvalue localization theorem stated in the supplement, Section 6.33.



(a) Compressed resolvent.

(b) First-order resolvent.

(c) Copy leakage.

(d) First-order eigenvalue localization.

Figure 3: Log-log finite-M checks. The compressed resolvent slope tracks the M^{-1} rate of Theorem 3.2; the first-order resolvent, squared copy leakage, and first-order eigenvalue localization track the M^{-2} rates of Theorems 3.5 and 3.10 and the first-order eigenvalue localization theorem stated in the supplement (Section 6.33). Fitted slopes and additional rate diagnostics are reported in the supplement, Section 7.

The remaining numerical checks – boundary transport heatmaps, chirality flip, multiboundary component drifts, flux insertion, collar/contour/refinement diagnostics, weak-disorder boundary transport, and the nonconstant positive copy-side confinement – are all collected in the supplement, Section 7. A symbolic verification appendix for the local crossing calculation supporting Theorem 5.5 is given in the supplement, Section 6.9.

7 Conclusion

We constructed a boundary model for the universal Chern Hamiltonians by replacing a triangulated surface with boundary by its closed double and adding a large positive potential on the reflected side. The relative fundamental class of $(\mathcal{K}, \partial\mathcal{K})$ gave an intrinsic Poincaré–Lefschetz boundary Hamiltonian on the original triangulation. Compressing the doubled Hamiltonian to the original side recovered this relative Hamiltonian up to a boundary-collar operator, and the jammed Hamiltonian converged to the compressed boundary operator in the large-jam limit by quantitative Schur-complement estimates.

Three buckets of results. For an external reader auditing this paper, we partition the results into three buckets according to which external inputs each one uses. We are explicit because the headline integer winding has different dependency status in different cases, and a single label could obscure that.

(A) *Fully internal results (no external theorem).* The finite-dimensional boundary theory (intrinsic Poincaré–Lefschetz Hamiltonian, collar comparison, and Schur-complement estimates of Theorems 2.8, 3.2, 3.5 to 3.7, 3.10 and 3.11); the internal bulk spectral equivalence (Theorem 3.16); the periodic triangular benchmark, namely the exact-determinant gap (Theorem 5.2), the isolated-crossing Chern calculation (Theorem 5.5), and the integer Toeplitz winding in the straight periodic case (Theorems 4.10 and 5.9). These results are proved in the paper without invoking any external theorem and without invoking the Ewert–Meyer Mayer–Vietoris boundary identification.

(B) *Internal modulo the Higson–Prodan closed-bulk theorem.* The Roe-algebra K_1-class identity $\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M}; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))) \in K_1(\mathcal{I})$ of Theorem 4.9 relies on Higson–Prodan's closed-bulk gap and Chern theorem [8, 9], imported as Theorem 4.2 of Section 4. Apart from this single external input, every step of Theorem 4.9 is proved internally; the hypothesis check that doubled-and-jammed families satisfy HP's hypotheses is Theorem 4.3.

(C) *Conditional on an external integer pairing.* The general-partition integer winding ± 1 statement (Theorem 4.11) is conditional on the existence of an integer-valued Chern-jump-compatible pairing between $K_0(\mathcal{A}/\mathcal{I})$ and $K_1(\mathcal{I})$. Such a pairing exists internally for the periodic Toeplitz case (already in bucket (A)); for general partitions the construction of the integer pairing is not carried out in this paper. This is the only result whose integer value depends on an external construction beyond the HP closed-bulk theorem, and we flag it explicitly as conditional.

In particular: the integer winding ± 1 for the straight periodic triangular interface is in bucket (A) (no external theorem at all); the K_1-class identity for general HP-covered boundary families is in bucket (B); and the integer evaluation of that K_1-class outside the periodic Toeplitz case is in bucket (C). No claim in this paper is named in a way that elides these distinctions; Theorem 4.10 carries the qualifier "in the Toeplitz periodic case" and Theorem 4.11 carries the qualifier "conditional integer winding for general partitions".

Beyond the HP-covered and periodic classes treated here, one possible extension is to prove uniform Fermi gaps for additional closed doubled refinement families. A second direction is to construct the integer Chern-jump-compatible pairing of bucket (C) for natural classes of non-Toeplitz partitions, which would promote Theorem 4.11 from bucket (C) to bucket (B). A third direction is to extend the internal boundary-map identification to curved or componentwise pairings that separate the contributions of individual boundary components.

Code and data availability

The companion numerical code and the online supplement are available at the project repository, which is hosted on GitHub: https://github.com/yspennstate/Doubling-and-Jamming-A-Universal-Chern-Hamiltonian-on-Triangulated-Surfaces-with-Boundary. The online supplement is also distributed through that same GitHub repository. The periodic triangular Bloch certificate script (`periodic_triangular_bulk_certificate.py`) and the companion numerical pipeline that generates the figures and tables of the supplement are included.

The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [36].

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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